NEXUS-SQL: Conversational NL-to-SQL with Adaptive Hybrid Semantic Retrieval and Resource-Aware Optimization

Poreddy Lokesh Reddy
Department of Computer Science and Engineering
SRM Institute of Science and Technology
Chennai, India
p.krishnalokeshreddy@gmail.com

Venkatesh C
Department of Computer Science and Engineering
SRM Institute of Science and Technology
Chennai, India
venkatec4@srmist.edu.in

Vismay Sajith
Department of Computer Science and Engineering
SRM Institute of Science and Technology
Chennai, India
vismaysajith4@gmail.com

Aaditya Gupta
Department of Computer Science and Engineering
SRM Institute of Science and Technology
Chennai, India
ag6457@srmist.edu.in

***Abstract*—NEXUS-SQL presents a conversational framework for interacting with structured databases through natural language while supporting semantic retrieval and adaptive query processing. The framework combines conversational context, ambiguity resolution, natural language to SQL generation, vector based semantic search, and hybrid retrieval within a unified workflow. Structured conditions are processed through SQL, while semantic requirements are handled through embedding based retrieval and combined using adaptive ranking. The framework further incorporates incremental execution error recovery to improve query reliability and employs resource aware optimization to control storage, latency, indexing, compression, and retrieval effort. An embedding evaluation process is used to examine semantic retrieval quality across diverse scholarly queries. The overall system is designed to provide accurate, efficient, and explainable database retrieval while allowing retrieval effort to adapt to query complexity and resource constraints.**

***Index Terms*—Natural Language to SQL, Conversational Databases, Vector Retrieval, Hybrid Retrieval, Embeddings, Query Optimization, Semantic Search, Error Recovery**

I. Introduction

A. Background

Modern database systems provide powerful structured querying but often require knowledge of schemas and query languages. Natural language interfaces reduce this barrier through automated Text-to-SQL generation, while schema-aware methods improve query construction over complex databases [1]–[3]. Conversational systems further extend database interaction by preserving context across multiple turns, while dense embeddings enable semantic retrieval beyond exact lexical matching [4]–[7].

B. Problem Statement and Motivation

A single natural language request may combine structured filters, semantic intent, temporal conditions, ambiguity, and conversational context. Recent studies address these challenges through decomposition, rewriting, execution feedback, hallucination mitigation, and ambiguity-aware processing [8]–[12]. However, selecting an appropriate retrieval strategy while controlling computation, latency, and storage remains an important systems challenge. Hybrid structured and vector retrieval provides a basis for combining exact constraints with semantic matching [13], [14].

C. Research Contributions

The main contributions of NEXUS-SQL are:

- Conversational SQL, vector, and hybrid retrieval.
- Query-adaptive routing with rank-based fusion.
- Incremental SQL error recovery.
- Embedding evaluation with robustness and compression analysis.
- Resource-aware control of search, storage, and latency.

The remainder of this paper presents the proposed architecture, embedding methodology, adaptive retrieval, error recovery, resource optimization, and experimental evaluation.

D. Research Gap and Distinguishing Contribution

Recent research has substantially advanced Text-to-SQL, conversational querying, ambiguity handling, semantic retrieval, hybrid search, and execution-based correction. Dynamic-SQL, for example, combines dense–sparse hybrid retrieval with schema linking and iterative execution feedback, while recent conversational benchmarks highlight the difficulty of resolving multifaceted ambiguity [12], [15]. These developments show that individual components of the NEXUS-SQL pipeline are established research directions rather than isolated new techniques.

The distinguishing contribution of NEXUS-SQL is their coordination at the query level under explicit resource constraints. The framework jointly selects SQL, vector, or hybrid retrieval, adapts retrieval effort using confidence and query complexity, applies rank-based fusion, and controls search depth, indexing, and vector representation according to accuracy, latency, and storage requirements. This provides a unified experimental framework for studying the relationship between semantic retrieval quality and resource consumption rather than optimizing these factors independently [13], [16], [17].

II. Related Work

A. Natural Language to SQL

Text-to-SQL converts natural language into executable SQL and relies heavily on schema linking and constrained generation. Spider introduced a major cross-domain benchmark, while RAT-SQL and PICARD improved schema-aware encoding and constrained decoding [1]–[3]. Recent methods explore data synthesis, decomposition, rewriting, hallucination mitigation, and execution-aware refinement [9], [10], [18]–[20].

B. Conversational Querying and Error Correction

Conversational Text-to-SQL preserves context across multiple turns, as demonstrated by SParC and CoSQL [4], [5]. Recent work further addresses ambiguous and unanswerable requests [11], [12]. SQL reliability has been improved through constrained generation, self-correction, error identification, execution feedback, and execution-driven bootstrapping [8], [21]–[24].

C. Embedding and Semantic Retrieval

Dense embeddings represent queries and documents in a shared semantic space, enabling retrieval beyond exact lexical matching. Sentence-BERT and dense passage retrieval established widely used approaches, while BEIR and MTEB provide comprehensive evaluation benchmarks [6], [7], [25], [26]. HNSW and related approximate nearest-neighbor methods improve retrieval efficiency for large vector collections [27].

D. Hybrid Retrieval

Structured SQL provides precise filtering, whereas vector search captures semantic relationships. Recent systems such as ACORN and Semantic SQL combine structured predicates with vector operations [13], [14]. Recent schema-linking methods also investigate stronger semantic bridging between user queries and database structures [28]–[30]. Reciprocal Rank Fusion provides an effective method for combining independently ranked candidate sets [31].

E. Vector Compression and Adaptive Optimization

Vector compression reduces storage at the cost of possible retrieval degradation. Product quantization and recent quantization methods investigate this quality-efficiency trade-off [32]–[34]. Adaptive indexing further shows that vector-search configurations can respond to workload and performance requirements [16]. Recent studies also examine efficiency trade-offs in complex Text-to-SQL processing [17].

F. Research Gap

Prior work has substantially advanced Text-to-SQL, conversational querying, semantic retrieval, hybrid search, error correction, and vector optimization. However, these capabilities are still commonly evaluated as separate components or under fixed retrieval conditions. Recent systems demonstrate increasingly strong individual capabilities in schema linking, correction, conversation, and hybrid retrieval [12], [22], [28]. NEXUS-SQL instead focuses on their query-level coordination through adaptive routing, rank-based fusion, confidence-driven escalation, early termination, and resource-aware control of retrieval effort.

III. Proposed NEXUS-SQL Architecture

NEXUS-SQL integrates conversational query understanding, structured SQL retrieval, semantic vector search, and adaptive resource control within a single query-level workflow. The system first interprets the query and its conversational context, then selects SQL, vector, or hybrid retrieval. Candidate results are ranked and evaluated for confidence, with additional computation performed only when required.

A. Conversational Query Understanding

Previous turns are retained to resolve follow-up queries and incomplete references. Temporal, metric, and contextual ambiguities are detected before query generation; clarification is requested when available context is insufficient [4], [5], [11].

B. Query Complexity and Adaptive Routing

Query complexity is estimated from filtering, semantic, temporal, ambiguity, and conversational features:

$$C_q = f(F_q, S_q, T_q, A_q, H_q) \tag{1}$$

The retrieval path is then selected as

$$P^* \in \{\mathrm{SQL}, \mathrm{Vector}, \mathrm{Hybrid}\}. \tag{2}$$

SQL is used for explicit relational constraints, vector retrieval for semantic requirements, and hybrid processing when both are present. This query-level routing forms the basis for controlling retrieval effort and resource consumption.

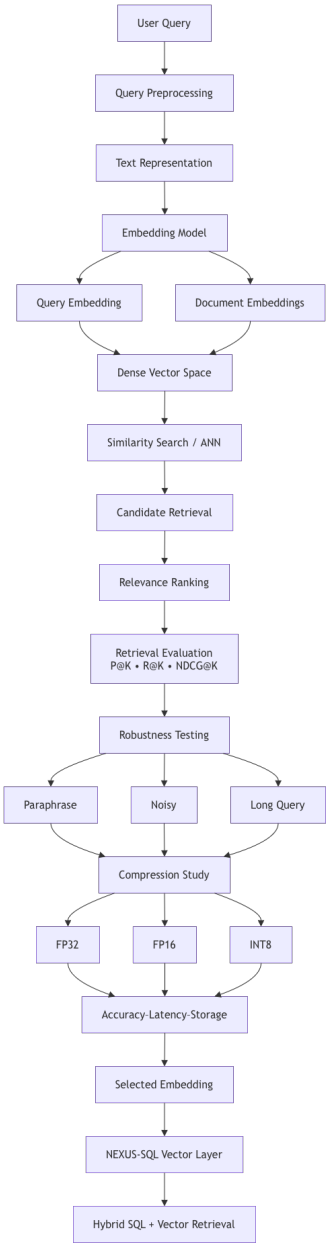


Fig. 1. Overall architecture of the proposed NEXUS-SQL framework.

C. Hybrid Retrieval and Adaptive Search

SQL and vector retrieval generate independent candidate sets that are combined using Reciprocal Rank Fusion [31]. Result confidence is estimated from ranking separation, agreement, and constraint satisfaction. High-confidence queries are returned directly, whereas low-confidence queries receive additional search depth or candidates. Retrieval terminates when further search provides insufficient improvement, reducing unnecessary computation [16].

IV. Embedding Model and Semantic Retrieval

A. Embedding Representation and Corpus

NEXUS-SQL maps scholarly queries and documents into a shared dense vector space. Document representations use titles, abstracts, keywords, research areas, and selected metadata. The corpus is normalized, deduplicated, and validated before embedding generation [6], [7], [35].

B. Model Selection and Evaluation

Multiple pretrained embedding models are evaluated using a common corpus and query benchmark. Selection considers retrieval quality, robustness, encoding cost, dimensionality, and storage requirements [7], [26]. Development, validation, and independent test sets are maintained to prevent test-set leakage.

C. Query Benchmark and Metrics

The benchmark covers exact, semantic, paraphrased, multi-constraint, conversational, ambiguous, noisy, and long-form queries, including hard negatives. Retrieval quality is measured using Precision@K, Recall@K, and NDCG@K:

$$P@K = \frac{|\mathrm{Rel} \cap \mathrm{Ret}_K|}{K}, \qquad R@K = \frac{|\mathrm{Rel} \cap \mathrm{Ret}_K|}{|\mathrm{Rel}|}. \quad (3)$$

Encoding time, retrieval latency, memory, and storage are also measured [7], [26].

D. Robustness and Compression

Robustness is evaluated using paraphrases, terminology variation, noisy inputs, long queries, and domain variation. Full-precision and compressed vectors are compared using retrieval quality, storage, and latency [32], [33].

V. Adaptive Hybrid Retrieval

NEXUS-SQL combines structured and semantic retrieval according to query requirements. SQL handles explicit constraints, while vector search captures semantic intent; queries containing both use the hybrid path [1], [13], [14].

A. Query-Adaptive Fusion

SQL and vector candidates are combined using Reciprocal Rank Fusion [31]:

$$RRF(d) = \sum_{m \in M} \frac{w_m}{k + r_m(d)} \quad (4)$$

where $r_m(d)$ is the rank of document d, w_m is the method weight, and k is the smoothing constant. The relative retrieval contribution is adjusted according to query characteristics:

$$S(d,q) = \alpha_q S_{SQL}(d,q) + (1-\alpha_q) S_{Vector}(d,q). \quad (5)$$

Algorithm 1 Incremental SQL Error Recovery

```
Generate and validate SQL
Execute query
if execution fails then
    Classify error
    Locate affected component
    Apply targeted correction
    Validate and re-execute
end if
Return result or failure explanation
```

B. Confidence and Adaptive Search

Confidence is estimated from ranking separation, retrieval agreement, and constraint satisfaction. High-confidence results are returned directly, while low-confidence queries receive additional candidates or greater search depth. Retrieval is terminated when the marginal gain

$$\Delta R_K = R@K - R@(K/2) \tag{6}$$

falls below the selected threshold, reducing unnecessary computation while preserving retrieval quality [16].

VI. Incremental SQL Error Recovery

Generated SQL is validated for syntax, schema consistency, safety, and semantic validity before execution. Execution failures are classified and the affected query component is repaired using available schema information [3], [8], [20].

A. Incremental Correction

NEXUS-SQL applies targeted correction to schema references, filter values, and join paths while preserving valid query components. This reduces unnecessary query regeneration after execution failures.

Performance is measured using recovery success rate, execution accuracy, correction iterations, and additional latency.

VII. Resource-Aware Optimization

NEXUS-SQL maximizes retrieval quality under explicit latency and storage constraints:

$$\max A \quad \text{s.t.} \quad L_{P95} \le L_{budget},\ S \le S_{budget} \tag{7}$$

where A denotes retrieval accuracy, L_{P95} tail latency, and S storage usage.

A. Adaptive Retrieval Policy

The system estimates query cost and selects SQL, vector, or hybrid processing. Search depth, candidate count, index configuration, and vector representation are adjusted according to query complexity and resource limits [16], [27], [32], [33].

$$Cost(p) = \lambda_1 \hat{L}_p + \lambda_2 \hat{C}_p + \lambda_3 \hat{S}_p - \lambda_4 \hat{A}_p \tag{8}$$

The selected policy is

$$p^* = \arg\min_p Cost(p). \tag{9}$$

Algorithm 2 Resource-Aware Retrieval

```
Estimate query complexity and cost
Select SQL, Vector, or Hybrid
Allocate search budget
Generate and rank candidates
if confidence is low then
    Increase search effort and re-rank
end if
if marginal gain is below threshold then
    Stop retrieval
end if
Return optimized result
```

TABLE I
Embedding Retrieval Performance

Model	P@10	R@10	NDCG@10
Word 1–2 Gram	0.0283	0.0283	0.0296
Word 1–3 Gram	0.0430	0.0430	0.0452
Character 3–5 Gram	0.0330	0.0330	0.0323
Selected Model	0.9312	0.9316	0.9320

Low-confidence queries receive additional search effort, while retrieval stops when the marginal gain becomes insufficient.

VIII. Experimental Setup

Experiments are conducted in a local Python and Oracle Database environment. Scholarly records, metadata, and vector representations are stored for structured and semantic retrieval.

The corpus is normalized, deduplicated, and validated before embedding generation. Multiple pretrained embedding models are evaluated using a common benchmark, while approximate nearest neighbor retrieval is configured with controlled search parameters [16], [27].

The benchmark covers exact, semantic, paraphrased, multi-constraint, conversational, ambiguous, noisy, and long-form queries. SQL-only, vector-only, static hybrid, adaptive hybrid, compressed vector, and complete NEXUS-SQL configurations are compared.

Performance is measured using Precision@K, Recall@K, NDCG@K, execution accuracy, recovery success, P50, P95, P99 latency, storage, and resource consumption.

IX. Results

A. Embedding Performance

The selected embedding configuration achieved a retrieval score of 0.931 on the evaluated benchmark. The reported embedding results are shown in Table I.

B. Retrieval and Adaptive Performance

SQL retrieval supports explicit constraints, while vector retrieval captures semantic intent. Hybrid retrieval combines both paths, with adaptive routing selecting the appropriate strategy and increasing retrieval effort for low-confidence queries.

C. Compression and Latency

Compression is evaluated by jointly considering retrieval quality, storage consumption, and latency. Tail latency is reported using P50, P95, and P99 to capture both typical and high-latency cases.

D. Error Recovery

Incremental SQL recovery is evaluated using correction success, execution accuracy, correction iterations, and additional latency.

E. Overall Performance

The complete NEXUS-SQL configuration is evaluated across retrieval quality, execution reliability, latency, and storage. The final configuration is selected according to the defined resource constraints.

X. Ablation Study

The major NEXUS-SQL components are evaluated by removing individual components and comparing their effect on retrieval quality, efficiency, and execution reliability.

A. Retrieval and Resource Components

Adaptive routing, rank fusion, confidence escalation, early stopping, and compression are evaluated against the complete configuration to assess their individual contributions [16], [31]–[33].

B. Conversation and Error Recovery

Context-aware processing and incremental SQL correction are independently disabled to measure their effects on conversational interpretation and execution reliability.

C. Score Normalization

Softmax normalization is evaluated only as an optional ablation and retained if it provides measurable improvement without significant computational overhead.

XI. Discussion

The selected embedding configuration provides effective semantic matching for scholarly queries, while hybrid retrieval is useful for requests combining structured constraints and semantic intent [6], [13].

Adaptive routing, confidence-based escalation, and early stopping reduce unnecessary retrieval effort while preserving retrieval quality. Compression introduces a trade-off among storage, latency, and retrieval performance [16], [32], [33].

Incremental SQL correction improves recovery from execution failures without requiring complete query regeneration. The modular design also allows alternative embedding models, indexes, routing policies, and compression methods to be evaluated within the same framework.

XII. Limitations and Future Work

Performance depends on embedding quality, corpus coverage, hardware, and external knowledge sources. Rule-based components may require adaptation for unseen queries. Future work will explore learned routing, domain-adapted embeddings, resource optimization, multimodal retrieval, and distributed vector search.

XIII. Conclusion

NEXUS-SQL integrates conversational querying, semantic embeddings, structured SQL, hybrid retrieval, incremental error recovery, and resource-aware optimization. The evaluation jointly considers retrieval quality, execution reliability, latency, and storage, providing a modular framework for efficient conversational database retrieval.

References

[1] T. Yu, R. Zhang, K. Yang, M. Yasunaga, D. Wang, Z. Li, J. Ma, I. Li, Q. Yao, S. Roman, Z. Zhang, and D. Radev, "Spider: A large-scale human-labeled dataset for complex and cross-domain semantic parsing and text-to-SQL task," in *Proceedings of the 2018 Conference on Empirical Methods in Natural Language Processing*. Brussels, Belgium: Association for Computational Linguistics, 2018, pp. 3911–3921.

[2] B. Wang, R. Shin, X. Liu, O. Polozov, and M. Richardson, "RAT-SQL: Relation-aware schema encoding and linking for text-to-SQL parsers," in *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*. Online: Association for Computational Linguistics, 2020, pp. 7567–7578.

[3] T. Scholak, N. Schucher, and D. Bahdanau, "PICARD: Parsing incrementally for constrained auto-regressive decoding from language models," in *Proceedings of the 2021 Conference on Empirical Methods in Natural Language Processing*. Online and Punta Cana, Dominican Republic: Association for Computational Linguistics, 2021, pp. 9895–9901.

[4] T. Yu, R. Zhang, M. Yasunaga, Y. C. Tan, X. V. Lin, S. Li, H. Er, I. Li, B. Pang, T. Chen, E. Ji, S. Dixit, D. Proctor, S. Shim, J. Kraft, V. Zhang, C. Xiong, R. Socher, W. Lasecki, and D. Radev, "SParC: Cross-domain semantic parsing in context," in *Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics*. Florence, Italy: Association for Computational Linguistics, 2019, pp. 4511–4523.

[5] T. Yu, R. Zhang, H. Er, S. Li, E. Xue, B. Pang, X. V. Lin, Y. C. Tan, T. Shi, Z. Li, Y. Jiang, M. Yasunaga, S. Shim, T. Chen, A. Fabbri, Z. Li, L. Chen, Y. Zhang, S. Dixit, V. Zhang, C. Xiong, R. Socher, W. Lasecki, and D. Radev, "CoSQL: A conversational text-to-SQL challenge towards cross-domain natural language interfaces to databases," in *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing*. Hong Kong, China: Association for Computational Linguistics, 2019, pp. 1962–1979.

[6] N. Reimers and I. Gurevych, "Sentence-BERT: Sentence embeddings using siamese bert-networks," in *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing*. Hong Kong, China: Association for Computational Linguistics, 2019, pp. 3982–3992.

[7] N. Muennighoff, N. Tazi, L. Magne, and N. Reimers, "MTEB: Massive text embedding benchmark," in *Proceedings of the 17th Conference of the European Chapter of the Association for Computational Linguistics*. Dubrovnik, Croatia: Association for Computational Linguistics, 2023, pp. 2014–2037.

[8] M. Pourreza and D. Rafiei, "DIN-SQL: Decomposed in-context learning of text-to-SQL with self-correction," in *Advances in Neural Information Processing Systems*, vol. 36, 2023.

[9] W. Mao, R. Wang, J. Guo, J. Zeng, C. Gao, P. Han, and C. Liu, "Enhancing text-to-SQL parsing through question rewriting and execution-guided refinement," in *Findings of the Association for Computational Linguistics: ACL 2024*. Association for Computational Linguistics, 2024.

[10] G. Qu, J. Li, B. Li, B. Qin, N. Huo, C. Ma, and R. Cheng, "Before generation, align it! a novel and effective strategy for mitigating hallucinations in text-to-SQL generation," in *Findings of the Association for Computational Linguistics: ACL 2024*. Bangkok, Thailand: Association for Computational Linguistics, 2024, pp. 5456–5471.

[11] M. Dong, N. A. Kumar, Y. Hu, A. Chauhan, C.-W. Hang, S. Chang, L. Pan, W. Lan, H. Zhu, J. Jiang, P. Ng, and Z. Wang, "PRACTIQ: A practical conversational text-to-SQL dataset with ambiguous and unanswerable queries," in *Proceedings of the 2025 Conference of the Nations of the Americas Chapter of the Association for Computational Linguistics: Human Language Technologies*. Albuquerque, New Mexico: Association for Computational Linguistics, 2025, pp. 255–273.

[12] T. Sarwar, F. Moghimifar, C. D. V. Hoang, X. Ma, S. C. Xu, F. Saleh, P. Zaremoodi, A. Sil, and K. Kirchhoff, "CLARITY: A framework and benchmark for conversational language ambiguity and unanswerability in interactive NL2SQL systems," in *Proceedings of the 64th Annual Meeting of the Association for Computational Linguistics*. San Diego, California, USA: Association for Computational Linguistics, 2026, pp. 1218–1250.

[13] L. Patel, P. Kraft, C. Guestrin, and M. Zaharia, "ACORN: Performant and predicate-agnostic search over vector embeddings and structured data," *Proceedings of the ACM on Management of Data*, vol. 2, no. 3, p. 120, 2024.

[14] A. Mittal, A. Bheemreddy, and H. Tao, "Semantic SQL – combining and optimizing semantic predicates in SQL," *arXiv preprint arXiv:2404.03880*, 2024.

[15] H. Hao, X. Zheng, and X. Yu, "Dynamic-sql: An adaptive nl2sql framework with multi-path fusion reasoning and execution feedback correction," *Scientific Reports*, vol. 16, p. 17565, 2026.

[16] J. Mohoney, D. Sarda, M. Tang, S. R. Chowdhury, A. Pacaci, I. F. Ilyas, T. Rekatsinas, and S. Venkataraman, "Quake: Adaptive indexing for vector search," in *19th USENIX Symposium on Operating Systems Design and Implementation*. Boston, MA: USENIX Association, 2025, pp. 153–169.

[17] T. Parekh, E. Hofmann-Coyle, S. Wang, S. S. R. Kothur, S. Prasad, and Y. Chen, "PExA: Parallel exploration agent for complex text-to-SQL," in *Proceedings of the 64th Annual Meeting of the Association for Computational Linguistics*. San Diego, California, USA: Association for Computational Linguistics, 2026, pp. 564–594.

[18] J. Yang, B. Hui, M. Yang, J. Yang, J. Lin, and C. Zhou, "Synthesizing text-to-SQL data from weak and strong LLMs," in *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics*. Bangkok, Thailand: Association for Computational Linguistics, 2024, pp. 7864–7875.

[19] Y. Xie, X. Jin, T. Xie, M. Lin, L. Chen, C. Yu, L. Cheng, C. Zhuo, B. Hu, and Z. Li, "Decomposition for enhancing attention: Improving LLM-based text-to-SQL through workflow paradigm," in *Findings of the Association for Computational Linguistics: ACL 2024*. Bangkok, Thailand: Association for Computational Linguistics, 2024, pp. 10 796–10 816.

[20] Y. Wang, P. Liu, and X. Yang, "LinkAlign: Scalable schema linking for real-world large-scale multi-database text-to-SQL," in *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing*. Suzhou, China: Association for Computational Linguistics, 2025, pp. 977–991.

[21] B. Xu, S. Li, H. Jing, M. Du, H. Song, H. Wang, and Y. Xiao, "Boosting text-to-SQL through multi-grained error identification," in *Proceedings of the 31st International Conference on Computational Linguistics*. Abu Dhabi, UAE: Association for Computational Linguistics, 2025, pp. 4282–4292.

[22] G. Qu, J. Li, B. Qin, X. Li, N. Huo, C. Ma, and R. Cheng, "SHARE: An SLM-based hierarchical action correction assistant for text-to-SQL," in *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics*. Vienna, Austria: Association for Computational Linguistics, 2025, pp. 11 268–11 292.

[23] B. Zhai, C. Xu, Y. He, and Z. Yao, "Optimizing reasoning for text-to-SQL with execution feedback," in *Findings of the Association for Computational Linguistics: ACL 2025*. Vienna, Austria: Association for Computational Linguistics, 2025, pp. 19 206–19 218.

[24] J. Zhang, H. Yang, K. Miao, R. Zhang, R. Pi, J. Gao, and X. Zhou, "ExeSQL: Self-taught text-to-SQL models with execution-driven bootstrapping for SQL dialects," in *Findings of the Association for Computational Linguistics: EMNLP 2025*. Suzhou, China: Association for Computational Linguistics, 2025, pp. 24 305–24 326.

[25] V. Karpukhin, B. Oguz, S. Min, P. Lewis, L. Wu, S. Edunov, D. Chen, and W. tau Yih, "Dense passage retrieval for open-domain question answering," in *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing*. Online: Association for Computational Linguistics, 2020, pp. 6769–6781.

[26] N. Thakur, N. Reimers, J. Daxenberger, A. Rücklé, and I. Gurevych, "BEIR: A heterogeneous benchmark for zero-shot evaluation of information retrieval models," in *Thirty-Fifth Conference on Neural Information Processing Systems Datasets and Benchmarks Track*, 2021.

[27] Y. A. Malkov and D. A. Yashunin, "Efficient and robust approximate nearest neighbor search using hierarchical navigable small world graphs," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 42, no. 4, pp. 824–836, 2020.

[28] Z. Hao, J. Huang, S. Shi, R. Cai, and B. Xu, "GenLink: Generation-driven schema-linking via multi-model learning for text-to-SQL," in *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing*. Suzhou, China: Association for Computational Linguistics, 2025, pp. 29 892–29 905.

[29] Z. Wu, Z. Li, J. Zhang, Z. He, J. Yang, Y. Zhao, R. Fang, B. Wang, H. Xie, S. Song, and Z. Li, "UCS-SQL: Uniting content and structure for enhanced semantic bridging in text-to-SQL," in *Findings of the Association for Computational Linguistics: ACL 2025*. Vienna, Austria: Association for Computational Linguistics, 2025, pp. 8156–8168.

[30] G. Liu, Y. Tan, R. Zhong, Y. Xie, L. Zhao, Q. Wang, B. Hu, and Z. Li, "Solid-SQL: Enhanced schema-linking based in-context learning for robust text-to-SQL," in *Proceedings of the 31st International Conference on Computational Linguistics*. Abu Dhabi, UAE: Association for Computational Linguistics, 2025.

[31] G. V. Cormack, C. L. A. Clarke, and S. Büttcher, "Reciprocal rank fusion outperforms condorcet and individual rank learning methods," in *Proceedings of the 32nd International ACM SIGIR Conference on Research and Development in Information Retrieval*. ACM, 2009, pp. 758–759.

[32] H. Jégou, M. Douze, and C. Schmid, "Product quantization for nearest neighbor search," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 33, no. 1, pp. 117–128, 2011.

[33] J. Gao and C. Long, "RaBitQ: Quantizing high-dimensional vectors with a theoretical error bound for approximate nearest neighbor search," *arXiv preprint arXiv:2405.12497*, 2024.

[34] J. Gao, Y. Gou, Y. Xu, Y. Yang, C. Long, and R. C.-W. Wong, "Practical and asymptotically optimal quantization of high-dimensional vectors in euclidean space for approximate nearest neighbor search," *Proceedings of the ACM on Management of Data*, vol. 3, no. 3, pp. 1–26, 2025.

[35] L. Wang, N. Yang, X. Huang, B. Jiao, L. Yang, D. Jiang, R. Majumder, and F. Wei, "Text embeddings by weakly-supervised contrastive pre-training," *arXiv preprint arXiv:2212.03533*, 2022.